

## GUINAN, China

WMO: 529550

Lat: 35.5867N

Lon: 100.7400E

Elev: 10246

StdP: 10.01

Time Zone: 8.00 (E08)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -5.3       | -2.0 | -27.4                         | 1.8 | 6.2  | -23.6 | 2.2 | 10.8 | 14.8                  | 29.9 | 12.0 | 28.9 | 3.2                   | 160  | 0.622 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 20.3                   | 76.2            | 55.3 | 72.9 | 53.4 | 70.1 | 51.9 | 58.6                | 70.0 | 57.1 | 67.8 | 55.6 | 65.0 | 5.0                  | 140  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 55.0                            | 95.2 | 62.4 | 53.6 | 90.4 | 61.2 | 52.2 | 85.6 | 60.1 | 30.9          | 70.1 | 29.6 | 68.4 | 28.4 | 65.2 | 64.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| 12.5              | 10.4 | 9.0 | -11.9                      | 82.0 | 2.8                | 2.7 | -14.0                                              | 84.0 | -15.6      | 85.5 | -17.2      | 87.1 | -19.2      | 89.0 |  |  |
|                   |      |     | -13.2                      | 61.0 | 2.8                | 1.7 | -15.2                                              | 62.3 | -16.8      | 63.3 | -18.4      | 64.2 | -20.4      | 65.5 |  |  |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 38.9   | 16.1  | 23.3  | 32.1  | 42.0  | 47.9  | 53.7  | 58.0  | 57.9  | 50.5  | 39.8  | 27.3  | 17.8  |  |  |
|                                                                        | DBStd       | 15.66  | 6.14  | 7.04  | 6.50  | 6.10  | 5.26  | 4.49  | 4.73  | 5.03  | 5.10  | 5.40  | 5.73  | 5.08  |  |  |
|                                                                        | HDD50       | 4771   | 1051  | 746   | 555   | 248   | 103   | 14    | 1     | 2     | 52    | 317   | 682   | 1000  |  |  |
|                                                                        | HDD65       | 9525   | 1516  | 1166  | 1020  | 690   | 530   | 340   | 223   | 229   | 434   | 780   | 1132  | 1465  |  |  |
|                                                                        | CDD50       | 737    | 0     | 0     | 1     | 8     | 38    | 125   | 249   | 246   | 68    | 2     | 0     | 0     |  |  |
|                                                                        | CDD65       | 15     | 0     | 0     | 0     | 0     | 0     | 1     | 6     | 8     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 199    | 0     | 0     | 0     | 0     | 2     | 14    | 80    | 90    | 13    | 0     | 0     | 0     |  |  |
|                                                                        | CDH80       | 14     | 0     | 0     | 0     | 0     | 0     | 0     | 5     | 8     | 1     | 0     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 4.0    | 3.7   | 4.2   | 4.7   | 4.9   | 4.6   | 3.9   | 3.8   | 3.7   | 3.3   | 3.5   | 3.7   | 3.6   |  |  |
|                                                                        | PrecAvg     | 15.5   | 0.1   | 0.1   | 0.3   | 0.8   | 2.0   | 2.6   | 3.5   | 3.1   | 2.1   | 0.7   | 0.1   | 0.0   |  |  |
| (15) Precipitation                                                     | PrecMax     | 22.2   | 0.3   | 0.4   | 1.2   | 1.9   | 3.7   | 4.6   | 6.2   | 5.9   | 4.3   | 1.6   | 0.4   | 0.2   |  |  |
|                                                                        | PrecMin     | 9.6    | 0.0   | 0.0   | 0.0   | 0.0   | 0.3   | 1.3   | 1.7   | 1.5   | 0.7   | 0.0   | 0.0   | 0.0   |  |  |
|                                                                        | PrecStd     | 2.7    | 0.1   | 0.1   | 0.3   | 0.6   | 0.8   | 0.8   | 1.0   | 1.1   | 0.9   | 0.4   | 0.1   | 0.0   |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 46.1  | 53.5  | 63.0  | 69.2  | 72.5  | 76.1  | 80.1  | 80.7  | 75.0  | 64.1  | 53.3  | 43.3  |  |  |
|                                                                        |             | MCWB   | 28.7  | 32.3  | 37.7  | 42.2  | 46.0  | 53.5  | 57.4  | 56.8  | 54.0  | 44.5  | 33.6  | 27.5  |  |  |
|                                                                        | 2%          | DB     | 39.5  | 47.6  | 55.9  | 64.5  | 68.1  | 71.3  | 76.1  | 76.4  | 69.2  | 59.8  | 48.8  | 38.8  |  |  |
|                                                                        |             | MCWB   | 25.1  | 29.6  | 34.7  | 40.2  | 44.4  | 51.1  | 55.8  | 55.8  | 51.5  | 41.4  | 31.6  | 24.9  |  |  |
|                                                                        | 5%          | DB     | 35.1  | 43.1  | 51.0  | 60.5  | 64.4  | 68.2  | 72.7  | 72.7  | 65.1  | 56.1  | 45.6  | 35.2  |  |  |
|                                                                        |             | MCWB   | 22.7  | 27.5  | 32.0  | 38.8  | 43.7  | 49.8  | 54.5  | 54.2  | 49.1  | 39.5  | 30.1  | 23.2  |  |  |
|                                                                        | 10%         | DB     | 30.8  | 38.6  | 47.0  | 56.9  | 61.0  | 65.4  | 69.7  | 69.3  | 61.7  | 52.8  | 41.8  | 32.0  |  |  |
|                                                                        |             | MCWB   | 20.2  | 25.3  | 30.3  | 37.3  | 42.5  | 49.0  | 53.5  | 53.1  | 47.3  | 37.8  | 28.2  | 21.1  |  |  |
|                                                                        | 0.4%        | WB     | 28.8  | 32.4  | 38.1  | 45.0  | 50.6  | 59.1  | 61.9  | 60.3  | 56.8  | 49.0  | 34.8  | 27.9  |  |  |
|                                                                        |             | MCDB   | 45.3  | 51.8  | 60.0  | 60.2  | 63.0  | 69.9  | 73.4  | 73.3  | 67.7  | 59.2  | 50.7  | 42.1  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 25.2  | 29.8  | 35.3  | 42.1  | 48.3  | 55.7  | 58.9  | 58.4  | 54.5  | 44.7  | 32.4  | 25.2  |  |  |
|                                                                        |             | MCDB   | 39.0  | 46.7  | 52.8  | 58.9  | 60.2  | 64.6  | 70.1  | 70.2  | 65.0  | 53.5  | 47.0  | 38.1  |  |  |
|                                                                        | 5%          | WB     | 22.7  | 27.7  | 33.0  | 40.4  | 46.4  | 53.5  | 57.2  | 57.1  | 52.4  | 41.9  | 30.7  | 23.2  |  |  |
|                                                                        |             | MCDB   | 34.6  | 42.5  | 48.4  | 56.5  | 58.6  | 62.3  | 68.2  | 67.9  | 61.1  | 51.4  | 44.0  | 34.9  |  |  |
|                                                                        | 10%         | WB     | 20.3  | 25.4  | 31.0  | 38.6  | 44.7  | 51.3  | 55.7  | 55.7  | 50.4  | 39.4  | 28.7  | 21.2  |  |  |
|                                                                        |             | MCDB   | 30.3  | 38.1  | 45.4  | 53.4  | 55.9  | 60.4  | 64.9  | 65.2  | 58.0  | 49.0  | 40.8  | 31.5  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 29.7  | 28.8  | 27.1  | 25.0  | 21.8  | 19.9  | 20.3  | 21.4  | 20.9  | 24.4  | 28.8  | 29.8  |  |  |
|                                                                        |             | MCDBR  | 34.8  | 34.0  | 33.2  | 31.3  | 29.4  | 27.0  | 26.3  | 26.9  | 27.0  | 28.8  | 32.5  | 32.9  |  |  |
|                                                                        |             | MCWBR  | 23.3  | 20.6  | 17.6  | 13.9  | 12.0  | 10.8  | 10.2  | 9.9   | 11.4  | 13.6  | 18.2  | 21.9  |  |  |
|                                                                        |             | MCDBR  | 33.7  | 32.4  | 29.5  | 25.8  | 21.7  | 20.0  | 19.4  | 20.6  | 21.0  | 23.0  | 29.8  | 31.9  |  |  |
|                                                                        | 5% WB       | MCWBR  | 22.6  | 19.6  | 15.7  | 11.7  | 9.7   | 9.0   | 8.4   | 8.2   | 9.7   | 12.1  | 16.8  | 21.3  |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.228 | 0.266 | 0.320 | 0.365 | 0.370 | 0.361 | 0.348 | 0.341 | 0.316 | 0.281 | 0.240 | 0.217 |  |  |
|                                                                        | taud        |        | 2.499 | 2.385 | 2.198 | 2.093 | 2.182 | 2.306 | 2.406 | 2.420 | 2.463 | 2.491 | 2.521 | 2.550 |  |  |
|                                                                        | Ebn at Noon |        | 318   | 312   | 300   | 291   | 290   | 291   | 294   | 295   | 299   | 303   | 310   | 317   |  |  |
|                                                                        | Edh at Noon |        | 25    | 32    | 43    | 50    | 47    | 41    | 37    | 36    | 33    | 29    | 24    | 23    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1052  | 1292  | 1648  | 1909  | 1919  | 1857  | 1877  | 1783  | 1472  | 1342  | 1145  | 1000  |  |  |
|                                                                        | RadStd      |        | 66    | 64    | 82    | 87    | 100   | 145   | 184   | 135   | 115   | 90    | 51    | 30    |  |  |

## Historical Trends

|      |                        | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------|------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|      |                        |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                        | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) | Station Only           | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (3 neighbors) | +0.50 | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page